

A history of rationality in economics: a computational approach

Abstract [needed for conferences]

Rationality has been one of the most central and extensively debated concepts in economics since the early twentieth century. Its centrality partly stems from its multiple meanings and the diverse ways in which economists have employed the term. Given the long and complex history of such a widely used concept, bibliometric analysis and text mining offer valuable tools for gaining a clearer understanding of the evolution of its meanings and uses. Drawing on a large database of full-text economics articles (approximately 300,000), we propose the use of large language models (LLMs), such as BERT, to examine semantic shifts over more than a century of economic publications. LLMs also facilitate the identification of a corpus of texts centered on rationality, which can then be analyzed through bibliometrics and network analysis. Our objectives are threefold: (1) to map the different uses and meanings of the terms “rationality” and “rational” over time; (2) to trace the emergence of new uses and meanings; and (3) to identify the major intellectual influences shaping these new uses and meanings.

Long abstract [needed for conferences]

Rationality has been one of the most central and extensively debated concepts in economics since the early twentieth century. From decision theory, expected utility theory, to rational expectations, many subfields of economics have their specific history of a complex relationship with the concept of rationality. Beyond the inner contributions of the discipline, interdisciplinary exchanges and controversies between economics and other social sciences have often taken rationality as a central point of departure to discuss the epistemological boundaries of economics.

While we find a few existing attempts at writing the history of rationality in economics, it is a daunting task to tackle such a large object. Typically, existing histories are often fragmented and focus on one particular subfield in relation to rationality, such as the rise of rational expectations in macroeconomics or the emergence of behavioral economics with Kahneman and Tversky’s contributions. In addition, although such histories often identify clear milestones such as the publication of canonical articles, the exact timing and scale of these transformations often remain uncertain.

We provide a pedagogical methodological blueprint for using quantitative tools to tackle such historical themes. Our goal is to show historians of economics how these tools can help address their own research questions, but also how they may help open new lines of inquiry. For this purpose, we focus on the history of rationality in economics because its centrality captures the interest of many historians of different subfields of economics, but also because existing attempts at writing a general history of rationality in economics often proved underdeveloped in some aspects.

In the present article, we apply scientometric and natural language processing techniques to a large corpus of twentieth and twenty-first-century economic literature, mixing JSTOR

full-text data with Web of Science bibliometric data to study the history of rationality in economics. From simple techniques that historians can integrate into their everyday research to more advanced tools reserved for specialized investigations, we will tackle what quantitative approaches can and can't do for historians. Our analysis showcases how these methods can pinpoint the timing and scale of conceptual shifts, measure the influence of key texts, and map the relationships of different sub-fields of economics concerning rationality.

Introduction

Several essays have examined the emergence of quantitative methods in the history of economics (Cherrier and Andrej; Edwards et al., etc.). A common limitation of these programmatic contributions lies in the generality of their discussions. Given the virtually unlimited number of quantitative methods potentially relevant to historians of economics—as well as the wide array of research questions these methods can address—these reflections often remain abstract and offer little in the way of concrete or practical guidance. As a result, they tend to provide broad overviews without clearly articulating what quantification entails in practice, and how it can be both meaningful and challenging to integrate into historical research.

The purpose of our contribution: no programmatic or methodological discussion of methods. No proof of concept and use of methods to answer a specific question. Illustrating how these methods can be useful, and how they involve a series of questions. We propose a step by step approach of a very large topic (talking to many historians). Above all, our discussion aims at illustrating one of the main principle of the methods we use: they rely on a constant back-and-forth between quantitative results, various indicators, reading of the primary and secondary literature....

A literature review on the history of rationality

Rationality is perhaps one of the most central and encompassing concepts in economics, alongside “market” and “competition”. Several contributions have offered broad narratives tracing its evolution over several decades or within specific subfields of economics. Moreover, a substantial body of work by historians of economics has focused on specific concepts—such as the emergence of expected utility theory, the development of rational expectations, and the evolution of bounded rationality. Our aim in this section is not to provide an exhaustive summary of all historical contributions on the concept of rationality, but rather to highlight key developments and debates in this long and complex history.

The idea that economic agents behave in a rational way was already implicit in many classical arguments, such as the Ricardian theory of rent, where farmers rationally select land according to its marginal fertility and profitability. While it is probably vain to look for a single precise origin of the concept in economics, it certainly gained prominence with the rise of scientific and abstract reasoning in the late nineteenth century. The question of rational behavior was closely intertwined with the broader issue of the proper scope of economic analysis (Giocoli

2003). A well-known example is John Stuart Mill, who in *Essays on Some Unsettled Questions of Political Economy* wrote:

"What is now commonly understood by the term 'Political Economy' is not the science of specu-

With the marginalist the economic agent was increasingly conceived in terms of calculative ability—for instance, Jevons portrayed him as a calculating machine balancing pleasures and pains that would eventually be formalized by the hypothesis of utility maximization (Maas, 1999). From marginalist theory, therefore, a first meaning of rationality crystallized around the idea of utility maximization. Based on the fundamental psychological premise that agents are self-interested, rational behavior was understood as the consistent selection of those means that maximize utility given available constraints.

The Second World War and the Cold War was a critical turning point in the development of the concept of "rationality" (Erickson et al., Mirowski). The key institutions in the Cold War context and the emergence of the concept of "rationality" were the RAND corporation (Erickson et al) and the Cowles Commission (Herfeld). Both were central to the development of operations research and linear programming in the late 1940s, involving figures such as Jacob Marschak, Tjalling Koopmans or George Dantzig (Duppe and Weintraub 2014). These developments, in turn, played a central role in helping Kenneth Arrow and Gérard Debreu to establish the proof of the existence of a general equilibrium in an economy composed of rational agents (Duppe and Weintraub book; Kirtchik and Boldyrev). Through these contributions, the concept of rationality came to be distinguished from concepts such as "reason" (Erickson et al.) or "intelligence" (Klaes and Sent). Rationality emerged as a formal concept, independent of personality or context, materializing through a set of "rigid rules that determine unique solutions" (Erickson et al. 2013, 3). From this axiomatic tradition emerged a second meaning of rationality, centered on the idea of consistent choice: agents were considered rational insofar as their preferences—whatever their psychological origins—satisfied a set of formal conditions that guaranteed internal coherence (Giocoli 2003).

At the individual level, the 1940s also marked a turning point in the formalization and axiomatization of economic behavior, most notably through the development of expected utility theory by Oskar Morgenstern and John Von Neumann in *Theory of Games and Economic Behavior* (Morgenstern and Von Neumann, Leonard ...). Morgenstern and Von Neumann formulated a set of axioms governing individual preferences (transitivity, completeness, independence, etc...) which allowed for the derivation of a value function capable of ordering agents' choices (Moscati 2023). Leonard Savage (1954) extended this framework by developing subjective expected utility theory, which addressed decision-making under uncertainty—that is when objective probabilities are not known.

Such modeling quickly developed along two dimensions: a positive one, seeking to represent how agents behave in the "real world," and a normative one, concerned with identifying optimal procedures for decision-making (Herfeld, Hands). As rationality came to be redefined in terms of *consistent choice* rather than as an empirical description of the *Homo Oeconomicus*, the normative dimension expanded the discussion beyond individual behavior to encompass broader issues of social choice and the design of "rationalizing" policies.

Another key institution in the history of the concept of rationality in economics was the Graduate School in Industrial Administration at the Carnegie Institute of Technology (Klein 2015). With funding from the Office of Naval Research obtained in the 1950s, a team of GSIA economists—including Charles Holt, Franco Modigliani, John Muth and Herbert Simon—was tasked with focusing on the theory of the firm and improving methods for production and inventory planning. This project was pivotal in the emergence of both Simon’s concept of “bounded rationality” and Muth’s “rational expectations” (Sent, 2002). While Muth’s concept was initially intended for use in microeconomic analysis of firm behavior, it was soon adopted and further developed by Robert Lucas and Edward Prescott (1971), both at GSIA at the time, and by Thomas Sargent (Darity et al., Duarte, 2025). Rational expectations quickly became a highly debated concept in a variety of theoretical frameworks requiring forward-looking behavior—for instance, in macroeconomics where it became the foundation component of dynamic stochastic general equilibrium (Goutsmedt et al.; Sergi 2020) or in finance (Delcey & Sergi 2020). While the rational expectations hypothesis was gaining popularity after the 1970s, expected utility theories were, during the same period, increasingly challenged—particularly through critiques rooted in paradoxes of rationality raised by Maurice Allais (1953) or Daniel Ellsberg (1961; see e.g. Zappia 2018). These critiques notably contributed to the development of Prospect Theory by Daniel Kahneman and Amos Tversky (1979), and the subsequent rise of behavioral economics (Truc, ...). In this context, “bounded rationality” experienced renewed interest after the 1980s (Sent 2018).

This history of the different meanings and uses of the concept of rationality is also a history of changing methods. It reflects shifts in both mathematical techniques and conceptions of rationality itself: a passage from a “system-of-forces”—analyzing “economic processes generated by market and non-market forces” (Giocoli, 4)—to a “system-of-relations,” in which the “main subject is the investigation of the existence and properties of economic equilibria in terms of the validation and mutual consistency of given formal conditions (5). In this transformation, mathematical methods and tools, such as axiomatization and fixed-point theorems became central to the discipline of economics (Weintraub 2002, Herfeld 2017). Game theory also progressively found its place in economics, even though it remained marginal in the years immediately following Morgenstern and Von Neumann’s *Theory of Games and Economic Behavior* (Giocoli 2003, Erickson 2015). The history of rationality is also the history of new alliances between economics and engineering. The work of Holt, Modigliani, Simon and Muth facilitated the importation of control theory into economics (Klein 2015), while the new classical economics—Lucas, Prescott, and Sargent—introduced elements of “information engineering,” inspired by information theory (Boumans 2020). Finally, this history of the concept of rationality also intersects with the so-called “applied turn” in economics (Backhouse and Cherrier): challenges to expected utility theories coincided with an increasing reliance on experimental methods—an approach that became central to the development of behavioral economics (Truc...).

What can the use of quantitative methods contribute to this already rich historiography? Many contributions have focused on the pioneering works of scientists like Von Neumann, Koopmans or Arrow. However, as Giocoli (2003, PP) has emphasized, the fact that these foundational contributions were developed in the 1940s and 1950s does not imply that new

conceptions of rationality were immediately adopted. The timing and diffusion of such “innovations” remain complex issues in the history of science that we seek to investigate. Moreover, the exponential growth in publications after the 1970s has likely multiplied the meanings and applications of “rationality,” making it increasingly difficult to construct a comprehensive picture. We offer a broader perspective on the diverse uses and interpretations of the concept, beyond the foundational texts that explicitly engage with rationality. This challenge is all the more significant given the concept’s proliferation across subfields—including general equilibrium theory, game theory, macroeconomics, and behavioral economics—as well as areas where rationality plays a less direct but still influential role, such as finance, firm behavior, public economics, social choice and welfare economics, as well as political economy and institutional economics. In addition, the concept is tied to distinct methods, formalizations, and mathematical tools, further complicating efforts to map its evolution.

This article does not aim to provide a definitive history of rationality—far from it—but rather to offer a series of snapshots illustrating how quantitative methods can confirm established historical findings, challenge or refine others, and expand the scope of historical inquiry.

Which methods and which challenges ?

Historians of economics today have at their disposal an expanding set of quantitative tools—from, for instance, traditional statistical inference, machine learning, textual analysis, network analysis and bibliometrics. Recent discussions of the “quantitative turn” in the history and philosophy of economics have focused on weighing the pros and cons of these methods, stressing their complementarity with qualitative approaches, and speculating about the future of this relationship. Yet most of these reflections remain at a general level, offering broad overviews of techniques that risk being too abstract to give readers a concrete sense of what quantitative analysis actually entails in practice. We take a different path. In our view, what matters most is to examine what kinds of questions and challenges quantitative analysis raises at each stage of the research process. To do so, we ground the discussion in our own inquiry into the history of rationality in economics. This concrete focus allows us to emphasize (a) the crucial issue of sources and data; (b) how even simple quantitative assessments can be highly informative; (c) the various challenges and subjective choice involved in building and adapting tools; and (d) why the interpretation of results requires a careful study of various indicators and a fine-grained knowledge of both the corpus and its historical context. By following step by step the process of assembling and analyzing our corpus, we aim to provide both a practical example and a reflection on the broader methodological issues faced by quantitative historians.

From raw source to data

Discussions on quantitative methods often focus on the varieties of methods existing and their relationship to qualitative analysis. Yet, despite often relying on sophisticated techniques, analysis and interpretation are usually the last—and not always the most difficult—step. It is not unusual for most of the time to be devoted to data collection and wrangling, even if much of this work remains invisible in the final paper. Our corpus of interest is rarely available in a

ready-to-use database but rather as raw sources that need to be structured. In a nutshell, a quantitative historian must be as much a skilled data wrangler as a data analyst.

The first step is to find the data. In our case, we want a comprehensive collection of economic documents that may discuss the concept of rationality and we want to know how the concept is actually used in these discussions. A well-structured category of sources at our disposal are bibliometric databases that index research documents. One of their main advantages is that they provide relatively well-structured data, systematically categorizing key features of scientific output—such as authors, journals and affiliations, etc.. In addition, some of these databases record citation data, which enables the study of intellectual influence, diffusion patterns, and the structure of scholarly communication. One crucial missing element, however, is the text itself. If some textual features are available, such as the title and for the most recent periods the abstracts, the full text in which the discussion is actually taking place is generally missing or very costly. One exception is JSTOR that provides us X full texts of documents covering the period between 1900 and 2020.

Before proceeding, we step back to a methodological point: how the available data shape what we can ask—and answer—about rationality. The scarcity and structure of available data already affects our multiple stages of the research process, from the selection of research questions to the answers we provide. In practice, our research may be shaped as much by the availability of our data as by our own research preferences. In our case, Figures X show that our full-text database, like most available databases, disproportionately represents Anglo-Saxon economic journals, which are also those most systematically digitized and preserved. Not only do we tend to overlook other traditions of economic research, but this also raises a serious issue of presentism: the fact that these journals are visible in recent periods does not imply that they were equally central in earlier ones, nor that articles were the dominant vehicle for the diffusion of ideas. Rather, it reflects the fact that they had the resources and opportunities to digitize and maintain comprehensive digital archives of their publications. Moreover, several key economics journals are absent from JSTOR (e.g. *Ecological Economics*, *Energy Policy*, *Applied Economics*), while other bibliometric databases such as Web of Science or Scopus present their own limitations in coverage and accessibility¹

Once access to JSTOR full text is secured, another crucial step is to transform them into a usable database. In most cases, data are not readily available in a structured format; rather, they result from a process that involves cleaning, categorizing, and selectively removing or reformatting information from raw sources in order to produce a dataset suitable for computational analysis. This challenge is particularly true in the history of economics, where the primary sources are the texts themselves—often unstructured and interspersed with various layers of metadata, such as section headings, footnotes, bibliographic references, or editorial annotations.

[Figure X: image of raw text ?]

¹One of the main challenges for a quantitative history of economics is to move beyond reliance on proprietary digital libraries and to actively develop open, historically inclusive corpora. This includes incorporating materials produced in non-Anglophone countries and expanding the range of textual formats considered—such as books, book reviews, working papers, and conference proceedings.

We do not mean, however, to imply that data wrangling is a necessary but laborious step before “real” historical analysis begins. On the contrary, manipulating, cleaning, and structuring raw sources is both a fundamental and stimulating stage of the research process. As recalled by Lemerrier and Zalc (2018, 62), the collection and categorisation of sources is also “a moment to reflect on the sources and the purpose of the research.” It is a potentially creative moment in which the direct confrontation with raw data permits the reevaluation of initial hypotheses and the emergence of new questions. In our case, the cleaning and structuring of full-text documents raises many questions about what should be offered as input of our analysis, and consequently, what counts as a relevant economic text. Should we include book reviews, editorials, or working papers or conference proceedings? And how do we define what qualifies as an economic journal—should it be restricted to the traditional “core” journals, or extended to interdisciplinary ones where economics frequently appears? Even within a single article, the boundary of what may be the textual content to analyze is not always clear: should headings, abstracts, footnotes, appendix or acknowledgements be treated as part of the text or definitely excluded? Each of these decisions have important historiographical implications. It shapes the boundaries of our corpus and, ultimately, the history of rationality that our methods can reveal. There is rarely a single, definitive choice, but rather a series of trade-offs that must be made explicit

The first important choice we made was to restrict our analysis to English articles, because cross-language comparisons remain difficult. Cross-language textual analysis is challenging since most quantitative methods are designed to identify commonalities within texts that share a specific linguistic structure. Hence, the same topic discussed by two documents in two different languages will likely not be identified as related, since the language differences obscure their underlying semantic similarity. We also restrain our analysis to research articles and filter out book reviews, comments, editorial reports or obituaries. While some of these secondary documents may contain important information on the way rationality was debated, they are not sites of enunciation of new ideas within the discipline. Moreover, textual data are by nature heavy, and filtering documents make the overall corpus more tractable for large-scale computational methods. At the level of documents, we focused on the body text, removing peripheral elements such as acknowledgements, references, or appendices.

Studying Rationality through Texts

Once the corpus has been assembled, the next challenge is to identify how the concept of rationality is actually used in the texts. A straightforward first step is simply to measure the frequency of occurrence of the terms themselves. Figure X, for instance, shows the relative frequency (with respect to the total number of words published each year) of “rational” and “rationality.” The figure reveals a clear surge in the 1980s and 1990s, suggesting that discussions explicitly framed in terms of rationality became particularly prominent during this period.

But the key question is not only whether a document contains the word “rationality,” but rather what meaning the term takes in its specific context. In other words, we are less interested in mere keyword retrieval than in uncovering the different meanings and intellectual settings in which rationality is invoked. A simple way to begin addressing this question is through

co-occurrence analysis, that is, by looking at the neighboring words that most often appear alongside “rational” or “rationality.” Co-occurrences provide clues about the conceptual frames and debates in which the term was embedded. Figure X, for example, displays the five words that most frequently appear immediately before or after ‘rational’ and ‘rationality’ in each decade, highlighting how the associations of rationality shifted over time. Before the 1940s, the picture is rather heterogeneous: rationality is connected to a variety of terms, often closer to philosophy, psychology, or general notions of reasoning. From the 1940s onward, however, the emergence of choice theory brought rationality to the center of economic modeling, particularly as a way to describe and formalize human behavior. By the 1970s and especially the 1980s, the framework itself was being refined and challenged, with the appearance of notions such as rational expectations on the one hand, and bounded rationality on the other.

While co-occurrence analysis already provides insights into the evolution of rationality, it remains limited to a broad picture of the immediate lexical environment of a word. Such methods capture proximity in vocabulary, not necessarily proximity in meaning. For example, one sentence may speak of “rational behavior” while another refers to “profit maximization”: they share no terms, yet they clearly point to closely related conceptions of agency and choice.

To move from this broad picture to how each document actually mobilizes the concept—and to map semantic relationships that are not visible through shared keywords—we turn to recent advances in natural language processing, in particular to large language models (LLMs). These models represent text in ways that encode context and meaning, enabling us to compare sentences that are semantically similar even when their vocabularies diverge. Similarly, it allows us to see how the same word may have different meanings when its context of use changes. Although quantitative text analysis is not new, its scope has expanded with these methods. Before introducing our LLM-based approach, we take a step back to present the basic principles of text representation because these principles remain relatively unfamiliar in the history of economics.

One of the most crucial concepts in textual analysis is the text *representation*, that is how text units (words, paragraphs, sentences, etc.) are encoded into numbers that can be processed by a computer. The most basic and historically the most dominant representation was the bag-of-words representation. Imagine a corpus in which your unit of interest is the document. You may want, for instance, to gather similar documents together to make sense of your corpus. But how can you compare documents computationally, if texts are made of words and computers only process numbers? The* bag-of-words *representation provides a simple solution: it transforms each document into a vector of word counts. Each dimension of this vector corresponds to a specific word from the vocabulary, and its value reflects how many times that word appears in the document. Two documents that use many of the same words will have similar vectors that can be computed. This representation is, for instance, the representation used in the probabilistic topic models that became increasingly used in the history of economics (Ambrosino et al.; Truc and Goutsmedt, etc.).

Any representation transforms a rich source of natural language into a simplified numerical format and, inevitably, loses some of the complexity and nuance of meaning along the way. A crucial question, then, is: what kind of information is preserved, and what is lost? Because it focuses only on the frequency of words, the *bag-of-words* representation discards two essential

dimensions of language: context (i.e., the surrounding words) and word order. A document is represented as a bag in which words are blindly tossed. For instance, the two sentences “Hayek is right and Keynes is wrong” and “Hayek is wrong and Keynes is right” have exactly the same bag-of-words representation for a computer whilst having opposite meaning for humans.

A major breakthrough that overcomes these limitations has been the development of word embeddings (Mikolov et al. 2013). While the bag-of-words model represents texts as unordered vectors of word counts, word embeddings focus on representing *individual* words in a way that captures their semantic relationships. Embeddings represent each word as a point in a vector space, learned from patterns of word co-occurrence in a large corpora of texts.² Figure X illustrates this principle. We simulate word vectors in a two-dimensional space. In reality, embeddings exist in high-dimensional spaces, but the principle is the same: absolute positions are arbitrary, while relative distances encode semantic relationships. Words that are plotted closer are more similar in meaning, even if they rarely co-occur.

The first generation of embedding models, such as Word2Vec or GloVe, learned static word vectors: each word had a single position in the vector space, regardless of the sentences or documents in which it appears. This makes it more difficult to grasp differences in meanings of a same word. The more recent generation of embedding models, known as LLMs, builds on a different principle: rather than assigning a fixed vector to each word, they produce contextualized embeddings—that is, vectors that depend on the surrounding words (the context). In LLMs, the meaning of the word “rationality,” captured by a word embedding vector, may vary significantly depending on the contexts in which it is used in the studied corpus. Each vector captures an usage of rationality based on the contexts in which different authors wrote. In this way, ideas and conceptual transformations become measurable and quantifiable objects, opening new possibilities for the study of economic ideas.

We use Sentence-BERT (Reimers & Gurevych, 2019), a variant of the LLM BERT, designed to produce sentence embeddings rather than word embeddings. This model yields a single vector representation per sentence, which allows straightforward semantic comparison by measuring the similarity between sentence vectors. To illustrate how embedding vectors capture semantic information, we compare the real sentences from the database with a synthetic sentence: “*Economic agents are supposed to be rational.*” Table X reports the most similar sentences retrieved for various years.³

We vectorized more than 31 millions of sentences from the JSTOR database using Sentence-BERT. Our objective is to identify sentences that discuss rationality and related concepts. Rather than relying solely on the presence of the keyword *rationality*, we use cosine similarity between sentence vectors to detect semantic proximity. For instance, if sentence A explicitly

²The intuition behind word embedding is the distributional hypothesis, famously stated by the linguist John R. Firth (1957): “you shall know a word by the company it keeps”.

³One issue with using LLMs is that they are pre-trained on recent data, which raises the methodological problem of *presentism*. Contextualized embeddings partially mitigate this issue, as the vectorization of texts depends on the immediate linguistic context. As illustrated in Table 1, each historical period is associated with its own specific set of questions and, therefore, with particular contextual usages of the term *rationality*. However, the way these surrounding words influence the vector representation of *rationality* remains shaped by a bias toward present-day language and meanings. This limitation is discussed further in the conclusion.

mentions *rationality* and its vector is close to the vector of sentence B—which does not use the term—it is likely that B addresses a related idea.

A straightforward first approach would be to compute a single average embedding vector for all sentences containing the word *rationality* or *rational*, and to use this as a representative embedding for the concept. However, such an average would be present-biased: since the number of documents in the corpus grows sharply over time, the average would be disproportionately influenced by recent decades, thereby over-emphasizing contemporary meanings of *rationality* at the expense of older ones. To account for potential semantic shifts over time, we compute a moving average vector for each year, using a sliding window of five years (i.e., the year in question plus five years before and after). From 1900 to 2020, this method yields 120 yearly representative vectors, each reflecting the evolving contextual meaning of *rationality* in the economic literature. For each year, we then computed the similarity between sentences from the corpus and the respective representative vector and we kept the 1% closest sentences (X sentences).

We interpret these sentences as reflecting how economists mobilize the concept of rationality, in different historical contexts. Because of the huge number of sentences we have identified, we would like a method to group sentences using a similar meaning of rationality together. We then cluster the sentences using the K-means algorithm—one of the most widely used and straightforward methods for grouping LLM embeddings. Because the distribution of our identified sentences is present-biased (Figure C), we cannot run a clustering algorithm on the entire distribution without risking clusters being over-determined by recent periods. We therefore run a clustering algorithm separately for each decade since 1900 (merging the first two decades because of the small number of observations). Table 2 provides a concrete example of what we mean by a cluster. We selected two clusters—respectively from the 1980s and 1990s—and report the three sentences nearest to the representative vector for each cluster, along with metadata from the associated documents. Note that these textual clusters group sentences rather than documents; hence, a single document can, in principle, contribute to several meanings of rationality. Despite being arbitrarily separated by decade, it is clear that sentences in both clusters convey a very similar meaning of rationality centered on the concept of rational expectations. To capture this stability of meanings across time, we merge clusters between decades when these clusters are sufficiently close (i.e., in our case, with a cosine similarity superior to 0.7). Following this procedure, the two clusters presented in Table 2 are ultimately merged into a single, temporally continuous cluster. Figure X displays the complete set of results in the form of an alluvial plot.

To ensure the stability of our results, we compare them with a bibliometric analysis of the corpus. Such a comparison is not costless, but it is highly recommended for at least two reasons. First, crossing the results of advanced computational methods such as LLM embeddings serves as a form of robustness check.⁴ Second, the systematic comparison of sources and methods is a fundamental principle of historical analysis. As we already discussed,

⁴Sophisticated techniques used in the history of economics—whether topic modeling, network analysis, or contextual embeddings—often depend on modeling choices (for example, the number of topics in a topic model) that can substantially affect the output. Since these outputs are never self-interpreting and always require human qualitative assessment, it is always very costly to conduct robustness checks.

each source has its strengths and limitations, and confronting them against each other helps to overcome individual biases. For instance, we saw that our JSTOR full-text database, while comprehensive, does not index several crucial recent journals in economics—a gap that can be partly compensated by crossing it with citation databases. Moreover, while full texts provide invaluable insights into the meanings and contexts of concepts, citation data highlight patterns of direct intellectual influence and diffusion. On the one hand, we focus on the semantic uses of rationality and related discussions in the text themselves, independently of authors’ institutional proximity or their citation practice. On the other hand, with the bibliometric analysis, we look rather at how these ideas circulated and have been diffused across fields.

Studying rationality through citations

Beyond textual analysis, another important source of information are the features (or meta-data) of the documents (e.g., authors, institutions, citations). We focus here on citations to study how authors cite and are cited. While textual data focus on the instantiation of an idea, citations rather allow us to understand its lineages and diffusion. The evolution of an idea often depends as much on how it is circulated and appropriated by readers as on how it was first formulated by its authors.

One major issue for quantitative analysts using this method is temporal: since systematic citation practices are relatively recent, patterns of bibliographic coupling are difficult to interpret before the 1960s and practically impossible due to the poor quality of earlier data. In contrast to textual analysis, thus, bibliometric approaches are largely confined to the postwar period. A second limitation is again the scarcity and quality of data: categorizing references from thousands of documents is a complex, time-consuming, and often imperfect task—even if recent efforts have been made to develop tools in this direction (see, for instance, Goutsmedt anstyle [X]). This is why we rely on the Web of Science database, one of the best available sources of structured bibliometric data on publications and their references, although it is proprietary, costly, and not openly accessible. In addition to accessing citation data on most economic documents, we therefore augmented our initial JSTOR full-text database with citation data from Web of Science, as well as citation and abstract data from key journals missing in JSTOR, as discussed in Section #.

Citation data does not require sophisticated analysis to be relevant, and can be easily incorporated into the everyday toolset of historians of economics. Simply counting citations can be used to proxy engagement and better understand the impact of a particular author or publication. There are compelling reasons to incorporate citation studies in historical research. Beyond serving as a proxy for the extent to which ideas disseminate, highly cited papers are more visible and get more engagement (Matthew Effect), and some recent studies suggest that citations affect reading and citations patterns, as well as perception of quality (Teplitskiy et al., 2022): highly cited papers have a higher chance to be a “substantial intellectual influence”, and they are often read more thoroughly because they are perceived as of better quality.

Historians commonly discuss the impact, influence, and scientific recognition of the figures they study, often relying on proxies such as major grants, prizes, or honors. For instance, Sent (2004) traces the transition from the dominance of rational choice theory, to the failure of “old”

behavioral economics to bring psychology back in economics (e.g., Simon, George Katona), to the rise of “new” behavioral economics (e.g., Kahneman, Akerlof). Sent’s introduction and historical narrative is structured around pivotal moments such as major grants, Nobel Prizes, or the John Bates Clark medal. In this sense, citations can be seen as a complementary proxy—alongside these more traditional markers—for assessing impact and influence. In an extensive qualitative and quantitative study of the Nobel Prize in economics, Offer and Söderberg (2016) identify different groups among Nobel winners: some win the Nobel at their peak in terms of citations, but rapidly decline after. Others do not decline and are what they call “innovators with staying power”. Some are “still rising” winners, who receive the prize early in their career years before their peak, while a few are late winners who get the Nobel way after their peak influence. Placing these institutional rewards in relation to citation and publication patterns thus helps to clarify how recognition interacts with the longer processes of diffusion, reception, and appropriation of ideas.

To illustrate these points within our study of rationality, (Figure [BOUNDED]) computes the citation patterns of four seminal references in relation to bounded rationality in all economic journals and the top five. While the four articles chosen are partly arbitrary, they are four important publications discussed in the literature for different reasons: Allais (1953) and Simon (1955) are some of the earliest criticisms of neoclassical rational choice theory, which had both empirical and normative implications. Akerlof (1970) and Kahneman and Tversky (1979) are two of the earliest “new” behavioral economics publications, partly capturing the field’s emergence and subsequent development.

Examining the citation patterns of these four articles raise interesting elements in relation to several arguments prevalent in the literature. First, the influence of “new” behavioral economics dramatically overshadows that of “old” behavioral economics. This holds true both for the top 5 economic journals and all economics journals indexed in Web of Science. While Allais (1953) and Simon (1955) were never cited by more than 0.20% of all economics articles within the studied period, Kahneman and Tversky (1979) achieved at least 1% of all economics article citations by the late 2010s and their influence continued to grow. The difference in citation patterns informs us about both the scale of the differentiated influence of “old” vs “new” behavioral economics, but also the differentiated speed at which they gained acceptance. While the rise of the two “new” behavioral economics articles was rapid and relatively steady after their publication, Simon and Allais’ peak citations were around the time they received the Nobel, or way later in the 2010s. The peak citations around the Nobel are not surprising, as argued by Offer and Söderberg (2016), most Nobel recipients received a “Nobel premium”, that is a small bump in citations after receiving the Nobel. However, while Allais and Simon display this small bump, we can see that the case of Kahneman and Tversky (1979) is particularly extreme as after the nobel, his citation trend switched from decreasing to increasing with no stopping in sight over the period studied.

Looking at citations alone is very limited and can be expanded in a variety of ways. It is possible to study who cites a given articles (which disciplines, journal or sub-specialties of economics). It is also possible to explore the qualitative attributes of theses citations by looking at whether they are positive/negative, or how they are used (e.g., in introductions, as a methodological reference; see Cavalcanti et al., 2011 or Budi & Yaniasih, 2022 for examples).

Beyond simple citation counts, citations can also be used in more sophisticated and systematic ways to study relationships between authors or publications.

One prominent example is bibliographic coupling analysis, an increasingly used method in the history of economics that maps corpora of documents based on the proximity of their references. The central hypothesis of bibliographic coupling is that similarity in references reflects intellectual proximity between texts. Spatialized in two dimensions, texts appear as nodes, and the more references they share, the closer they are positioned. Mapping the relationship between authors and publications is an important tool to delineate fields and disciplines, but it can also reveal how they are structured—for example, according to a core-periphery pattern.

To run a bibliographic coupling analysis, one major issue lies in the delimitation of the corpus: which articles should be included in a corpus about rationality? A straightforward solution is to rely on keyword search queries; however, as we have already discussed, keyword-based approaches present important limitations: for instance, it is possible that an article on revealed preference or on the prospect theory does not mention often or at all the words “rational” or “rationality”. We use instead the features of our LLM model and JSTOR sentences embedding. For each article in our coupled JSTOR–Web of Science database, we compute the cosine similarity between its embedding—obtained by averaging all the article’s sentence embeddings—and the representative vector associated with its year of publication. We then retain the 10% of documents that lie closest in this semantic space, thereby selecting those most strongly engaged with the uses of rationality at this specific year. Because embeddings capture deeper patterns of meaning and association, we are able to identify documents that engage with debates related to rationality even when the word “rationality” itself does not explicitly appear in the text. This approach allows us to construct a bibliometric corpus that goes beyond keyword-based searches and traces the broader intellectual conversations surrounding the concept of rationality.

We then applied the dynamic network analysis developed by Goutsmedt and Truc (2022), which we present in detail in the appendix, that followed the same spirit as the method we developed for sentence clusters. Intuitively, this method does not produce a single network covering the entire period, but rather a series of overlapping networks built on smaller time windows to preserve historical accuracy. A clustering algorithm is applied to each of these networks, and clusters are subsequently merged across windows when they share a sufficient number of nodes (documents). In a bibliographic coupling analysis, a cluster is defined as a group of documents that share a significant proportion of references, thereby indicating intellectual proximity. Hence, our bibliometric cluster here reflects a group of documents that all discuss rationality and that tends to discuss it with similar references. While our LLM clusters capture semantic similarities at the level of sentences, our bibliometric clusters instead reveal intellectual lineages and shared citation practices at the level of documents.

After all these sophisticated analyses, however, an essential question arises: what is the place of qualitative readings and the close appreciation of our primary materials—texts and their ideas?

Do you still need to read ?

A recurring criticism of quantitative analysis is that it distances researchers from the economic ideas themselves. This view is misleading for several reasons. First, as we already discussed, data wrangling often requires a detailed engagement with texts. The very act of transforming a raw source into structured data is not given but entails many crucial decisions about cleaning, categorizing, or excluding materials. This requires a fine understanding of the corpus and a good knowledge of the context in which the texts were produced.

Second, no quantitative method provides objective, automatic, and ready-to-use outputs. Rather, the quantitative methods we use offer raw, uninterpreted statistical results that systematically require qualitative interpretation. For instance, in our analysis we use bibliometric and textual techniques to group texts together. This is in fact a classic strategy to reduce a large and complex set of texts to a manageable number of simple categories. But these clusters are constructed on the basis of statistical commonalities, whose actual conceptual meaning can only be characterized through qualitative assessment. More importantly, a good qualitative knowledge of the corpus and of intellectual debates at stake are indispensable to make sense of raw results. Thus, whether in the collection of data or their analysis, quantitative methods require a constant back-and-forth with qualitative reasoning.

This cross-referencing with previous readings, various indicators, and a selective examination of relevant articles is all the more essential given that the methods proposed here could be described as “unsupervised.” In other words, there is no a priori supervision or human labeling to guide the algorithmic classification; therefore, human interpretation remains central to evaluating the validity and usefulness of the output. Indeed, what is crucial in any application of quantitative methods is the careful “validation” of the results, whether through quantitative indicators or qualitative interpretation. In this context, we need to assess whether the categories produced by our methods—namely bibliometric intertemporal communities and meaning clusters—constitute a relevant and helpful classification of our corpus. A key issue here concerns the construction of indicators that help analysts decide which documents should be read as a priority. Any dashboard will depend on the corpus and the research context, but a sound strategy is to rely on a combination of indicators rather than on a single measure. It is also important to design indicators using information available from the historical period under investigation in order to avoid presentism. For instance, to understand a network community from a past period, it is tempting to look first at the most cited papers as a proxy for the community’s most important documents. Yet this can be misleading, since total citation counts reflect current popularity rather than what was influential at the time.

In the case of a large-scale study such as this one, most results are not surprising—and there is nothing wrong with that. On the contrary, given the large body of works that have contributed to the historiography of rationality, one should expect our method to confirm existing findings rather than to produce entirely new discoveries. Precisely because much of what can be uncovered through quantitative analysis has already, in some way, been identified by other means, these methods can be trusted. This confirmation provides the basis for using them to explore where the historiography is less developed. A couple of patterns from our results can be highlighted as confirmations of existing consensus in the historiography. The first example

concerns the story of how economics, through the development of rationality theory, distanced itself from psychology (Giocoli 2003). By analyzing a selected group of prominent economists from the first half of the twentieth century—such as Vilfredo Pareto, Irving Fisher, Lionel Robbins, and Paul Samuelson—Giocoli traces the long-standing debate on the relationship between economics and psychology. This central debate on the role of hedonism in explaining economic behavior is clearly reflected in Cluster 2 (Figure X, LLM alluvial), which extends from 1900 to the 1940s. The cluster gathers discussions between proponents of hedonistic foundations and their critics, such as institutionalists, who regarded this psychological grounding as dubious. For instance, in this cluster, one of the closest sentences to the representative vector of this is from Wesley C. Mitchell, who argued that “to find the basis of economic rationality in the development of a social institution directs our attention away from that dark subjective realm, where so many economists have groped, to an objective realm, where behavior can be studied in the light of the common day” (Mitchell 1916, 160).

The 1940s and the 1950s also revealed the existence of two textual clusters (which are relatively highly correlated together but not sufficiently to be merged according to our threshold), which displayed debates over economic freedom and free enterprise against economic planning. Interestingly, we find in these clusters various roundtables and special issues of essays discussing economic planning—see this roundtable (Perroux...) on Lange’s (1949) “The Practice of Economic Planning and The Optimum Allocation of Resources” (1949) involving François Perroux, Jan Tinbergen, Evsey Domar or Michal Kalecki— or “the proper spheres of individual freedom and collective control in the ‘good’ economy” (Taylor 1948)—in a collection of essays with among others Henry Simons. In these two clusters, Frank Knight appears as the most recurrent author.⁵ Somewhat linked to this is the textual cluster 53, which from the 1950s on, discussed the issue of social choices and welfare, with rationality being thought as a guide for decision makers.⁶

The postwar period also testified of intense debate about utility theory, in textual cluster 31 for the 1940s, with notably Friedman and Savage (1948), but above all after the 1950s with cluster 46. This cluster discusses the demand theory and questioned in the first year the use of cardinal utility and the possibility of intertemporal comparisons. Chicago economics like Friedman and Savage (1952), Stigler (1950) and, later on, Becker (1962) figure prominently in this group, while Nicholas Georgescu-Roegen (1954) proposed a stark criticism of the use of the concept of “utility” in economics, and the lack of a clear definition of its measurability.

For the most “recent” period, a broad overview of our results makes two very expected patterns. Whether in the bibliometric or textual analyses, after the 1970s, rational expectations became a central concept that occupies an important extent of discussion, represented by many textual clusters or bibliometric communities (see below). Similarly, after the 1980s, the debate and undermining of the standard rationality approach in economics, notably through the EUT, appeared recurrently in various textual clusters (textual clusters 83 or 93) that persisted until the end of our periods, while a clear behavioural economics cluster (106) appears in the 2000s. In the XXIth century, these thematics have become more prevalent than discus-

⁵See also textual cluster 31 on planning issues,

⁶Kenneth Arrow, Herbert Simon or James Buchanan are central in these debates.

sion on rational expectations. Something similar is visible through bibliometric analysis where a first community about “Behavioral Economics and Choice Theory” appeared in 1977-1984, and many communities then popped up to form a complex network of discussions around the modelling of rationality in economics after the 2000.⁷

Discussion

What can these complex quantitative methods and the resulting careful interpretation bring about the history of rationality? How may it help to enrich, complete and refine our understanding of the various and evolving meanings of rationality in economics? As this paper aims mainly at underlining concretely the interests, issues and uses of quantitative methods in the history and philosophy of economics, our goal is not to produce an alternative history of the concept of rationality, nor to produce a comprehensive depiction of its evolution.

Rather, we would like to highlight what appears to us as interesting pieces of results, confirming and strengthening some pieces of the literature in the history of rationality, broadening this literature, as well as, we hope, opening new paths and raising issues for further research.

Mixing the various indicators with qualitative analysis (reading in detail the different analyses produced from our methods and using the literature).

Our goal is thus also to illustrate how we “navigate” our results, cross different indicators, to produce interesting insights and narratives.

The stabilisation of the concept of rationality

Before diving into the interpretation of the clusters, the structure of the alluvial in Figure X (alluvial LLM) already reveals an important insight. In the first half of the 20th century, many clusters appeared and disappeared quickly, reflecting a certain instability. From the second half of the century onward, however, clusters become more stable and persist over time, indicating a clearer consolidation of meanings and uses.

A rational expectations revolution ?

One way to look at our results is to focus on one specific concept, more accurately formalised and with less possible meanings than “rationality”, i.e. “rational expectations”. The concept has an interesting early history: developed by John Muth in 1960 to account for price setting in agriculture, it gained popularity when being grasped by Muth’s colleagues at Carnegie Tech, Robert Lucas, Edward Prescott and Thomas Sargent, who applied it to macroeconomics, in a context far from Muth’s original application. That’s only in the early 1970s that rational expectations became an important concept in economics, soon becoming quite controversial, notably because of its consequences when applied to monetary or fiscal policies. At the end of the 1970s, the concept was also circulating in policy circles and newspapers, the latter claiming about an “Rational Expectations Revolution”.

⁷See for instance the communities “Behavioral Economics and Experiments”, “Ambiguity and Uncertainty”, “Intertemporal Choice and Control” or “State Preference Valuation”.

What our methods may be helpful to is for understanding the quick diffusion of the assumption in economics, beyond its controversial popularity linked to the debates over macroeconomic policies in a context of stagflation in the 1970s.

First, the results support, indirectly, that rational expectation gained popularity only in the 1970s, when applied to macroeconomics. Indeed, the concept of rational expectations is absent from the 1960s both in the bibliometric and textual results. As early as the 1970s, it occupied a central place in the literature dealing with rationality.⁸

The controversial dimension of rational expectations is visible first through text analysis. Two textual clusters about rational expectations emerged in the 1970s decade (Textual clusters n° 67 and 72). The first one is populated by articles discussing in-depth the rational expectations hypothesis.⁹ In this perspective, many articles are criticizing or questioning the theoretical or empirical relevance of rational expectations in the 1970s and 1980s (e.g.,...). Others are discussing the implication of the use of this hypothesis for various domains (examples...). The second one rather deals with the models using rational expectations, rather than the hypothesis in itself.¹⁰ Here again, in the first decades, in parallel with the promotion of the rational expectations model (Sargent is one of the major authors of the cluster), we find several criticisms of these models, for their incorporation of the hypothesis, but also for the other hypothesis used in complement. The best example is perhaps Ray Fair's (1978, 411) criticism of "one class of macroeconomics models that have recently been developed", these models relying on "(1) the assumption that expectations are rational, given the available information; (2) the assumption that information is imperfect regarding the current state of the economy; and (3) the postulation of an aggregate supply equation in which aggregate supply is a function of exogenous terms plus the difference between the actual and expected price level."¹¹ The large diffusion of is also visible through clusters that exist before rational expectations is used, but which then incorporate them, especially in the 1980s (ie the textual cluster 76 on monetary economics issues and the cluster 28, a cluster dating back from the 1940s, focusing on the theory of prices).¹²

⁸This reveals one important possible caveat of such methods: as they tend to display "significant" patterns, it does not work well for tracing and understanding carefully the origins of some concept or theory, as historical studies and archives can do.

⁹That's one of the strengths of our textual method: it does not simply identify sentences that mention the topic which is of interest to us (rationality, and more specifically, here, rational expectations); but it rather finds sentences which significantly *engage* the topic.

¹⁰It should be noted that this second textual cluster is not oriented as intensely on rational expectations than the first one; interestingly, the cluster aggregates various discussion on models (some on models of rational behavior in general, some about demand), but rational expectations models constitute an important part of these discussions.

¹¹This combination of defense of models with rational expectations and opposition continue in the next decades: for instance, in the 2000s and 2010s, we find sentences and papers related to DSGE models, as well as papers incorporating bounded rationality in macroeconomic models and defending behavioural macroeconomics.

¹²Other textual clusters linked to rational expectations are dealing with inflation expectations and term structure in the 1970s (textual cluster 78), or with forecasting issues (textual cluster 81), wage and unemployment (82), finance (87), the treatment of information (89) and econometric issues linked to the use of rational expectations (92). As soon as the 1980s also appear a quite theoretical cluster, crossing rational expectations and game theory (88).

Our bibliometric analysis offers a fine-grained approach to complement text analysis and to reveal some similar patterns but by other means: by focusing on referencing rather than textual use, and by relying on overlapping windows, it allows to observe with accuracy the emergence of new communities or the split of others. The main bibliometric community on rational expectations, which appears for the first time in 1966-1973, does not deal primarily with the treatment of the hypothesis in itself, but its implications for macroeconomic policy. Here, the controversy is not so much about the relevance of the hypothesis, but about its use in the modelling of the inflation-unemployment trade-off, and its consequences in terms of (in-)efficiency of monetary policy. Here, we retrieve the early opposition between the “old” and (future) “new” Keynesians (Goutsmedt et al., 2019).

The controversial aspect of rational expectations in the 1970s and 1980s likely helps to explain its “prevalence” of this community in our results. These controversies have been well covered by the literature on the subject (Ref). But, relying on our results, we can also argue that rational expectations have spread in economics through a process of diffusion by being applied to various economic problems. Relying on bibliometric analysis, this could be seen through two schematic types of diffusion. First, communities that were independent from the main rational expectations community (“Rational Expectations and Business Cycles”) started to use rational expectations. Let’s delve into the some communities that may represent these two types of diffusion process. Second, the appearance of new communities, which are outgrowing the main rational expectations community appearing in the 1970s. Here, the papers in these new communities are initially in the rational expectations community, and progressively autonomize, with new communities appearing, becoming autonomous from the initial one.

Regarding the first category, it’s interesting to see that actually, one community invested the issue of rational expectations well before the “Rational Expectations and Business Cycle” community. This community dealt with various subjects related to trade, demand or investment, but with a general equilibrium theory framework. Muth (1961)’s was part of this community. Later on it remains one of the important references of the community, even if rational expectations hypothesis was not totally central to it. In 1971-1978, this community becomes “Investment and Economic Growth”. In this community, Lucas’ first papers on investment played an important role (but these papers did not use rational expectations). Again, if rational expectations was not totally central to the community which mainly dealt with firms’ investment and adjustment costs, many significant references used rational expectations, like Cyert and DeGroot’s “Rational Expectations and Bayesian Analysis” (1974) or Townsend’s (1978). In 1975-1982, it became “Investment and Uncertainty”. This community welcomed Kydland and Prescott (1977, 1982) and Lucas (1978).¹³

Thus, beyond the well-known debates on business cycles and inflation, there was a community, not totally focused on macroeconomics issues, that existed in parallel in the 1970s. After the mid-1970s, we also find several trajectories of independent communities that start to employ more and more rational expectations. A first example appears in 1976-1983: two com-

¹³These articles received a good number of citations from the paper of this community, but they appear as “connector” in the sense that they were highly connected to nodes in other communities.

munities on finance topics (“Asset Pricing and Consumption” and “cl_319”) partially merged to form “Rational Expectations and Market Information”. At the heart of the community was the concept of imperfect information (Rothschild and Stiglitz, 1976, being an important node). A central paper was Grossman and Stiglitz (1980), taking inspiration from Lucas (1973) and the combination of rational expectations and noisy signals to criticize Fama and the possibility of efficient markets (see Delcey and Sergi, 2023). Similarly, a new cluster emerged in 1979-1986, “Game Theory and Information”, aggregating papers in game theory, now recurring more significantly to the concept of rational expectations. In this cluster, Kydland and Prescott (1977) or Barro and Gordon (1983) on rational expectations and time-inconsistency problems were major references, in parallel to Kreps and Wilson (1980) sequential equilibria model, or Selten’s (1975) refinement of Nash equilibrium. In the early 1980s, this community would split, with a new community emerging, “Monetary Policy and Inflation”, leveraging game theory to discuss how monetary policy can manage inflation, making concepts like credibility or reputation central.

A second process of diffusion involves the emergence of new communities around the rational expectations concept, but growing and separating from the initial main rational expectations community. In the early 1970s, an international macroeconomics community emerged, in great part from the “Rational Expectations and Business Cycles” community. If this community was focused on various international economics problem, the determination of exchange rate dynamics and the role of rational expectations hypothesis in this determination, notably with Dornbusch (1976), would quickly become central. From 1974-1981 on, another community grew out from the “Rational Expectations and Business Cycles” community: the “Inflation, Expectations and Interest Rates”, focusing on the interest rate term structure and the forecasting of inflation through interest rates. This community would constitute another landing zone for macroeconomics and finance, in which rational expectations and efficient markets will be close (Delcey and Sergi, 2023).

This issue of rational expectations diffusion would deserve a whole paper. The goal here was uniquely to show the diversity of trajectories and the fine-grained mapping that our analyses produce, opening possibilities of rich interaction between quantitative data, selective text reading and qualitative approaches. However, at the start of the 1980s, it is clear that half the networks was constituted by communities dealing with more or less intensity with “rational expectations”. At the same time, Behavioral economics has emerged as a community, but still represented a small part of the network, a situation that will change after the 1990s.

The Trajectory of Simon’s contributions and the birth of behavioural economics

It is well known that if Simon’s “bounded rationality” constitutes one of the most famous concepts related to rationality, and supported a significant criticism of the standard treatment of rationality in economics, its success was delayed in time, well after Simon’s (1955, 1959) article. However, our bibliometric and textual analyses allow us to explore in detail the trajectory of Simon’s concept.

For sure, Simon was an important character of the debates about rationality in the 1950s, and we can find his two seminal articles and his Nobel lecture (Simon, 1979) featured in good

places in almost a dozen of textual clusters (on social choice, firm behavior, price setting, etc...), testifying of how its discussion of economic rationality echoed a large specter of debates on this issue. However, if its contribution had already got the highest graal in economics in the late 1970s, his concepts of “bounded rationality”, “procedural rationality”, and “satisficing” remained marginal at this time. The Figure X above on Simon (1955) citations in economics testifies of its late reckoning, the article being increasingly cited after the 1970s, but only to grasp a significant share of citations after the 1980s.

The textual analysis offers perhaps an even more striking proof. Textual cluster 40, one of the biggest clusters of our textual analysis, aggregated over 70 years, from 1950 on, general discussion on rationality in economics. In the 1950s, if we find Simon (1955) as a recurrent article in this cluster, the discussion mainly turned around rationality in game theory—with contributions, for instance, by Thomas Schelling (1959) or Daniel Ellsberg (1956)—and expected utility theory—with contributions by Jacob Marschak (1950) and Herman Chernoff (1954). In the 1970s, the concept of bounded rationality has become the most prevalent one of the cluster, which also deals with “behavioral theory”. In the 1990s, the articles in the cluster, beyond bounded rationality, talked of agents “boundedly rational”, of “unbounded rationality” and “procedural rationality”.¹⁴ Clearly, Simon’s concepts have become central to this major cluster.

The bibliometric approach, with successive time windows, allows to follow closely even the smallest communities and brings complementary information. And that’s what allows us to follow the influence of Simon’s 1950s contributions, which appear quite reduced at first. In the first window (1960-1967), Simon (1959) is cited in a community with operational research at its heart (cl_5). In parallel to operational research, this community which persisted—but reduced in size—until 1969-1976, was progressively populated by organizational theory articles criticizing the hypothesis of firm’s profit-maximization in neoclassical models.. Two main examples here are Sydney Winter (1971), applying Simon’s satisficing principle to study firm’s behavior, or Richard Cyert (Simon’s colleague at Carnegie) and Kenneth George (1969). Another important actor of this literature was Harvey Leibenstein who proposed the concept of “X-efficiency”, criticizing neoclassical economics’s focus on allocative efficiency. Simon, Winter, Cyert, and Leibenstein constituted a relatively heterogeneous group of organizational theorists critique of the use of the rationality concept, notably through profit-maximization in economics.

After 1969-1976, this group will land in a smaller community (cl_175) which will be short-lived. References to Simon and this kind of criticisms of profit-maximization will remain coerced but will navigate through various small communities (cl_182, cl_251, cl_300 and cl_328). This instability underlined the marginal nature of this literature, often associated shortly with other types of contributions. In the final cluster of this bibliometric errand (cl_328), Simon’s and its fellow organizational theorists were now accompanied by another literature focused on firm theory, the one on transaction costs and property rights. Williamson (1979) was part of it, and Coast (1960) constituted an important reference. Eventually, this crossing of firm theory, critical of standard profit maximization, and institutional economics,

¹⁴Simon himself is among the most prevalent words of the cluster.

formed a now big community in 1977-1984: “Behavioral Economics and Choice Theory”. There, they were in company of Kahneman and Tversky (1979), in a community now aggregating various criticisms of the concept of rationality in economics. For a couple of years, Simon, Coase, Kahneman and Tversky were all important point of references in this major community.

From 1979-1986 on, a new specific “Behavioral Decision Theory” would born, coalesced around Kahneman and Tversky’s contributions. This community will remain until the end of our period, giving birth (as for rational expectations, see above) to several new autonomous communities, behavioural economics becoming one of the dominant fields of inquiry on rationality after the 2000s. As for references to Simon, they did not pop up any more as important references in any communities. We know from text analysis (see above) that its concepts of bounded and procedural rationality became regularly mentioned, but it had become relatively detached from its authors, and no bibliometric communities bear clearly his heritage.

While it has often been argued that the rise of “new” behavioral economics shadowed the pre-existing works of “old” behavioral economics, Simon (1955) has never been more cited than after the rise of “new” behavioral economics. This resurgence in citations allows for three interpretations. The first, a more positive one, suggests that Sent’s aspiration for a “revival of abandoned research directions” or Peter Earl’s (2022) wish for a reconciliation of old and new behavioral economics was realized, with “new” behavioral economics indeed stimulating a renewed interest for earlier ideas. A second, more critical interpretation suggests a form of intellectual appropriation. It posits that these “old” references were transformed and reinterpreted to align with the “new” behavioral economics agenda. Thus, the rise of behavioral economics may have renewed interest in older scholarship, but potentially through a biased and presentist lens. Finally, a third and the most critical interpretation posits that the citation increase is evidence of a growing backlash against NBE from an OBE standpoint. Scholars may be citing Simon and Allais to emphasize a core philosophical divergence. Finally, a third and even more critical view is that the rise in citations indicates a growing critique against “new” behavioral economics, articulated from the perspective of “old” behavioral economics.

While counting citations alone cannot settle which interpretation is correct, it can orient qualitative reading and suggest further possible quantitative work. Studying the citations patterns observed towards Simon’s and other old behavioral economists with specialized and more advanced quantitative tools could help differentiate the different way these references are used. As show here for Simon’s trajectory in the 1960s and 1970s, this could be done by looking at the journals, authors and communities citing these authors, and/or by combining textual and citations data to identify if citations to Simon are superficially used by “new” behavioral economics or used to leverage a criticism of “new” behavioral economics by “old” behavioral economists (see Cavalcanti et al., 2011 or Budi & Yantiasih, 2022 for example of positive/negative citations classifications).

Conclusion

- American and English oriented analysis
- Limits to LLMs: BERT and presentism

Notes

Appendix

References